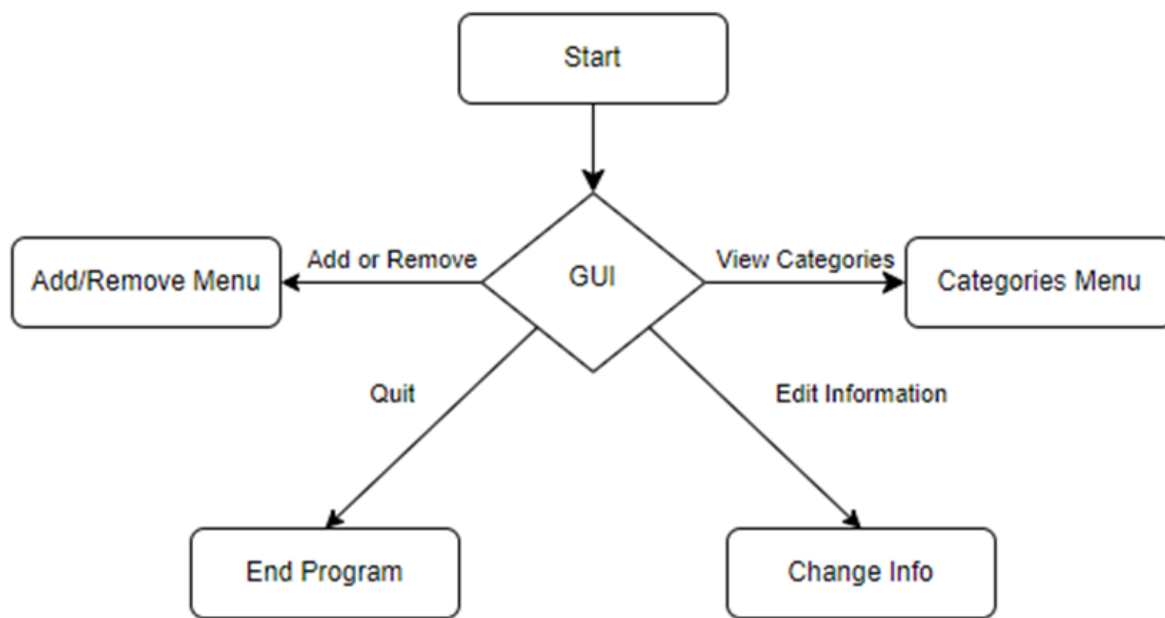


Criterion B: Solution Overview

Design Layout

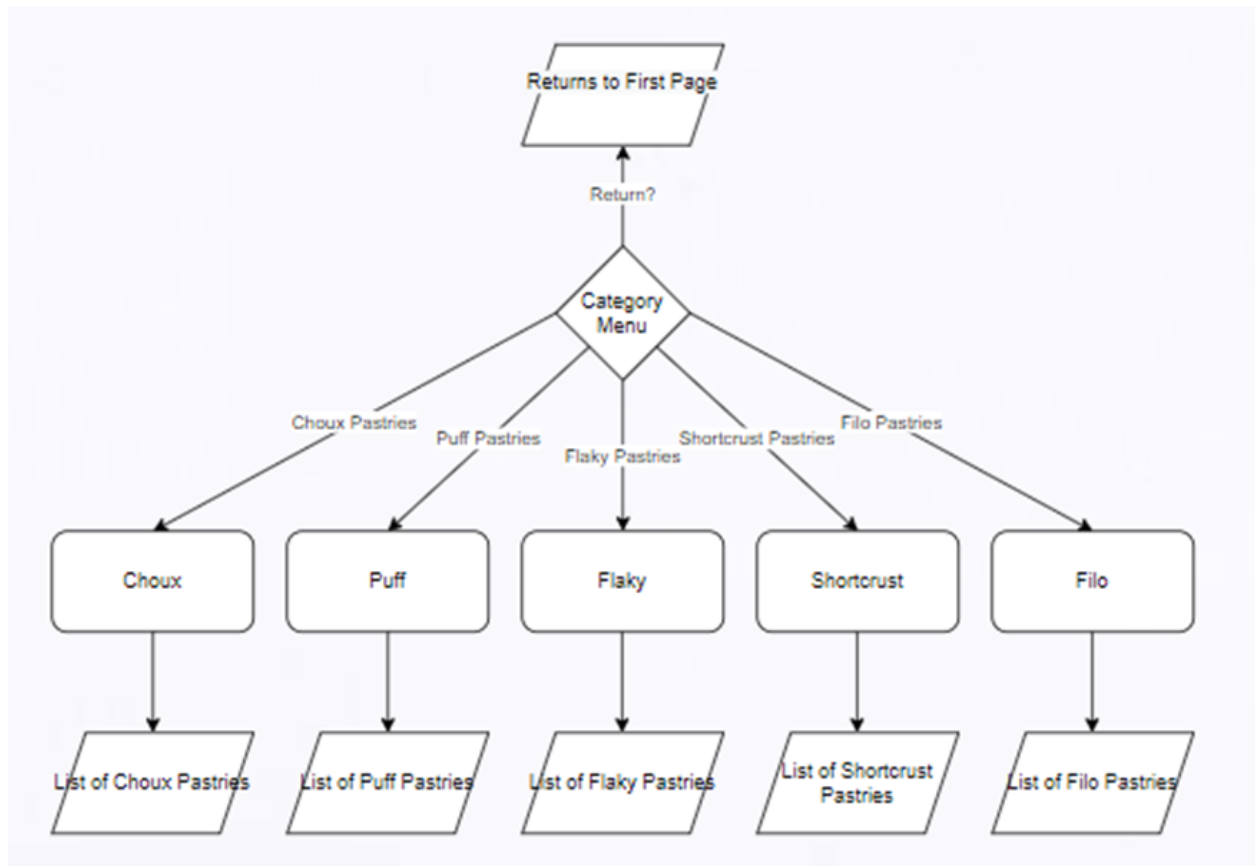
The design as a whole is:

Flow Chart 1:





Flow Chart 2A:



Pastry List

Choux Pastries

Puff Pastries

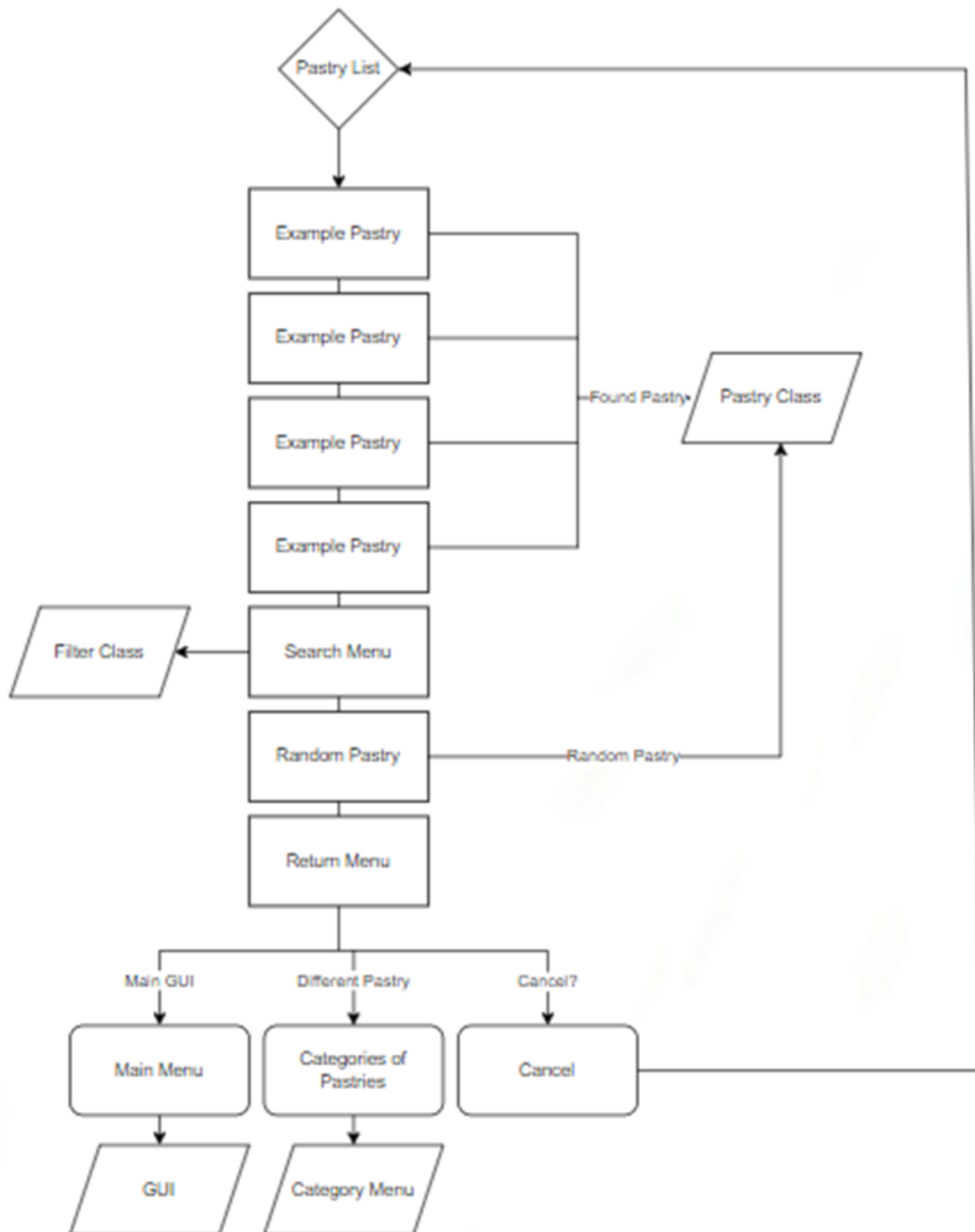
Flaky Pastries

Shortcrust Pastries

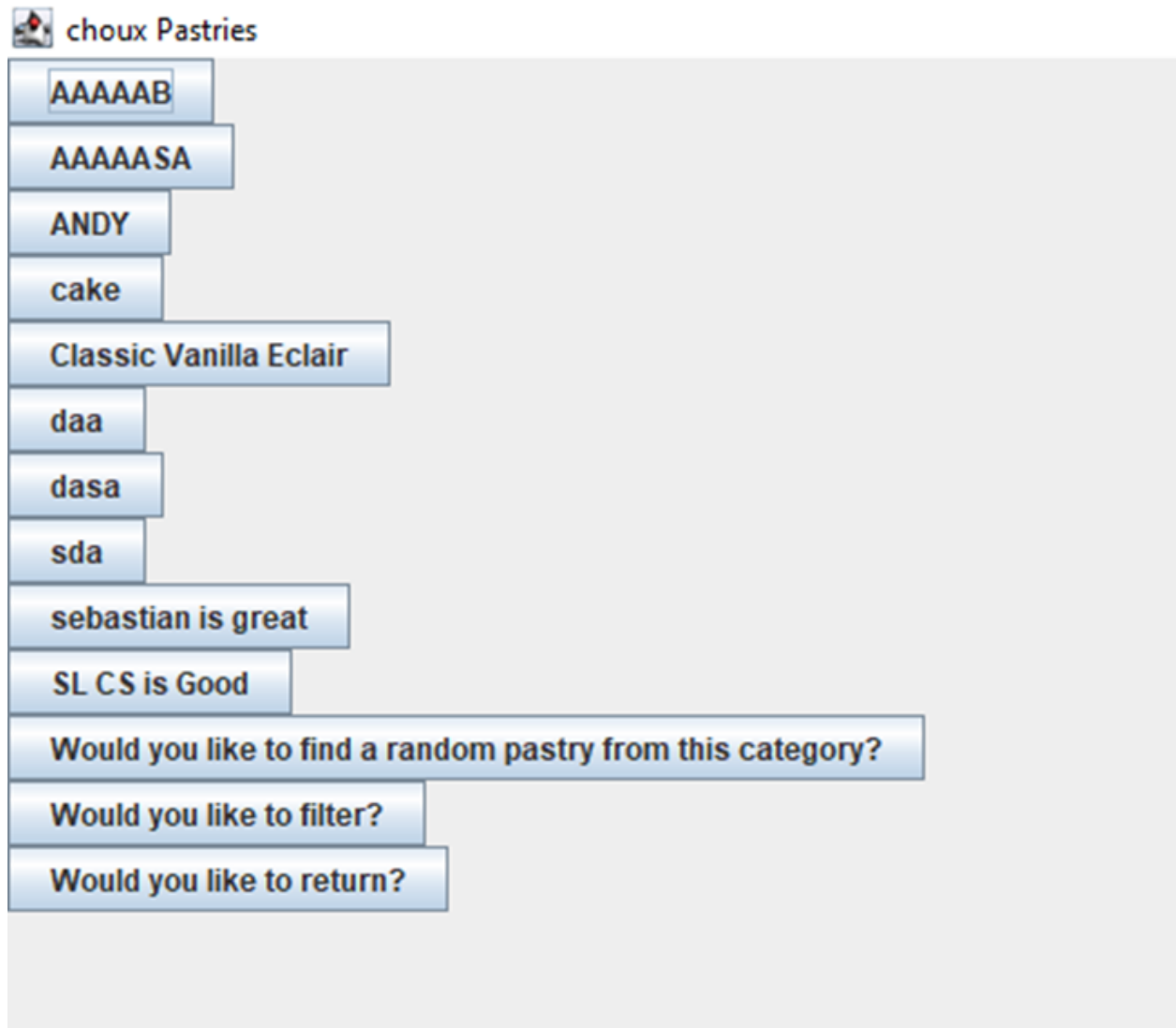
Filo Pastries

Return?

Flow Chart 2B:



The program looks under the folder with the pastries and takes the name of each file and puts them into a list then makes a button for each element of the list.



Return Menu displays another window over pastries. Depending on the button clicked, the button calls another class and method.



Return

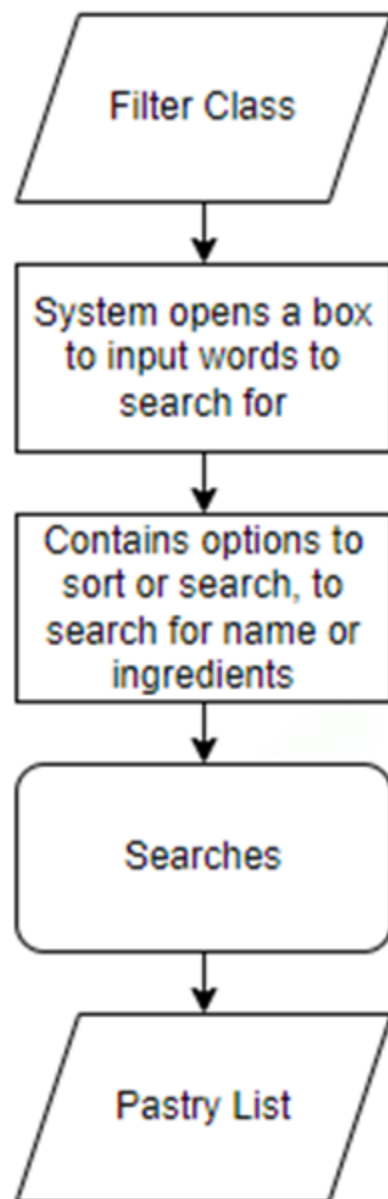
Which menu would you like to go to?

Main Menu

Categories of pastries

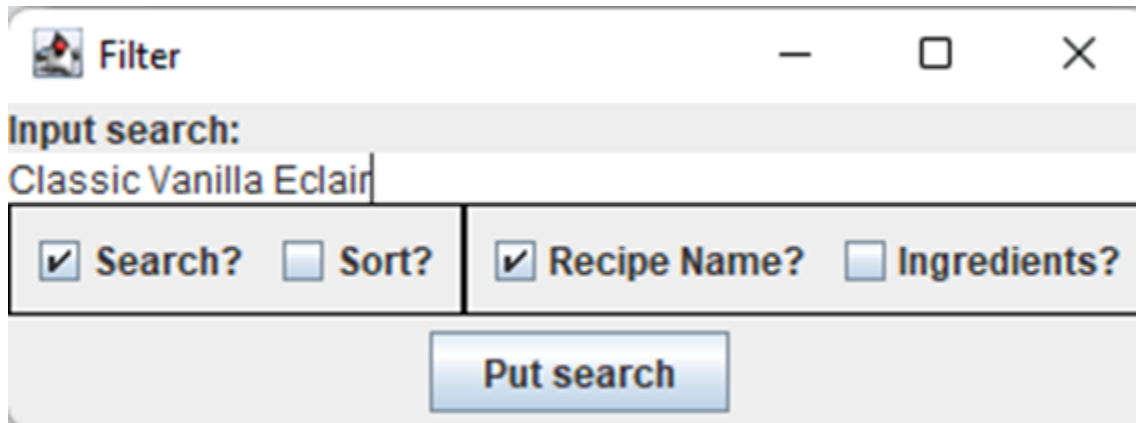
Cancel

Flow Chart 2C:



The Pastry List class gives the Filter class the list of pastries, for names, the program looks through the elements trying to find if the inputted search is found in any of the elements. For ingredients, the program reads the text file trying to find if the inputted word is under the list of

ingredients. If the program finds a pastry that meets the criteria, it is added to a new list. After going through all of the pastries, the program sends the modified list to the pastry list class.

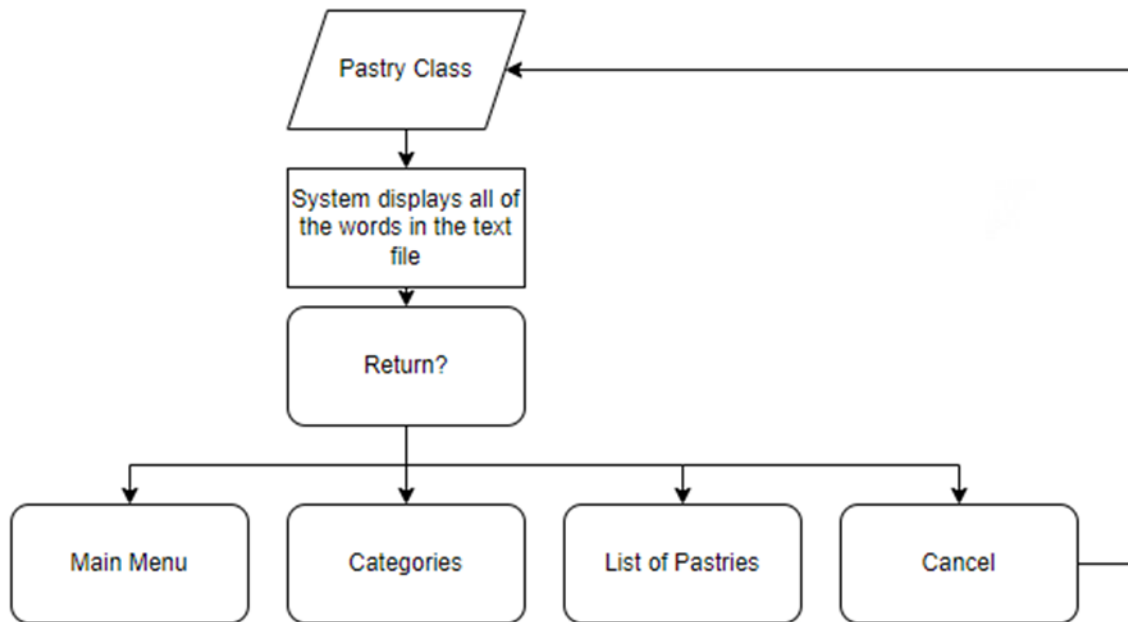


The screenshot shows a window titled "Filter" with standard Windows-style window controls (minimize, maximize, close). Inside the window, there is a label "Input search:" followed by a text field containing the text "Classic Vanilla Eclair". Below the text field, there are four checkboxes arranged in two groups. The first group contains "Search?" (checked) and "Sort?" (unchecked). The second group contains "Recipe Name?" (checked) and "Ingredients?" (unchecked). At the bottom center of the window is a button labeled "Put search".

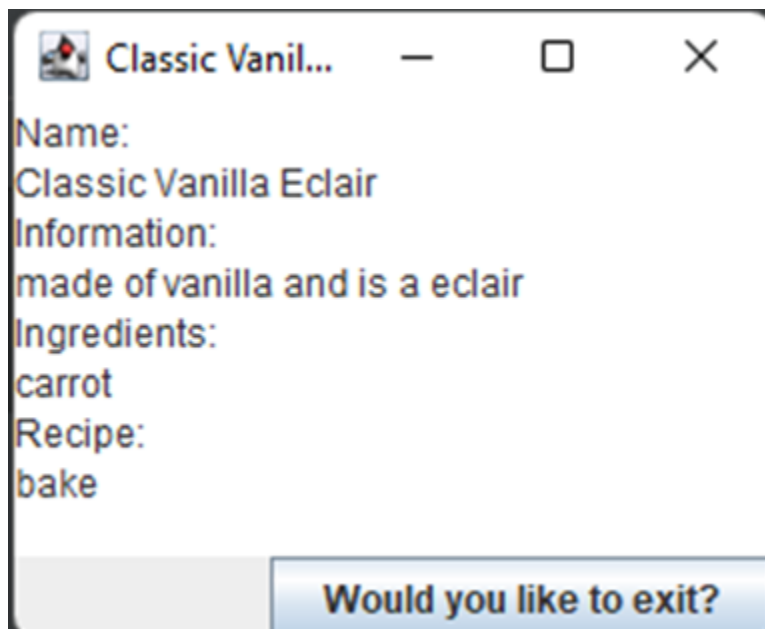


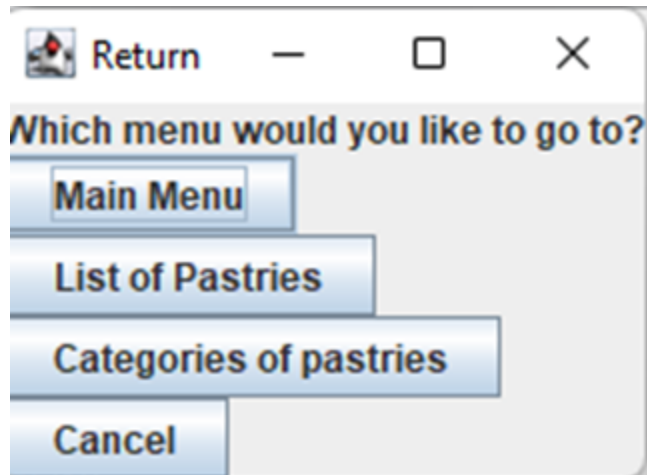
The screenshot shows a dialog box with a small icon in the top-left corner. It contains a text field with the text "Classic Vanilla Eclair". Below the text field are three buttons stacked vertically: "Would you like to find a random pastry from this category?", "Would you like to filter?", and "Would you like to return?".

Flow Chart 2D:

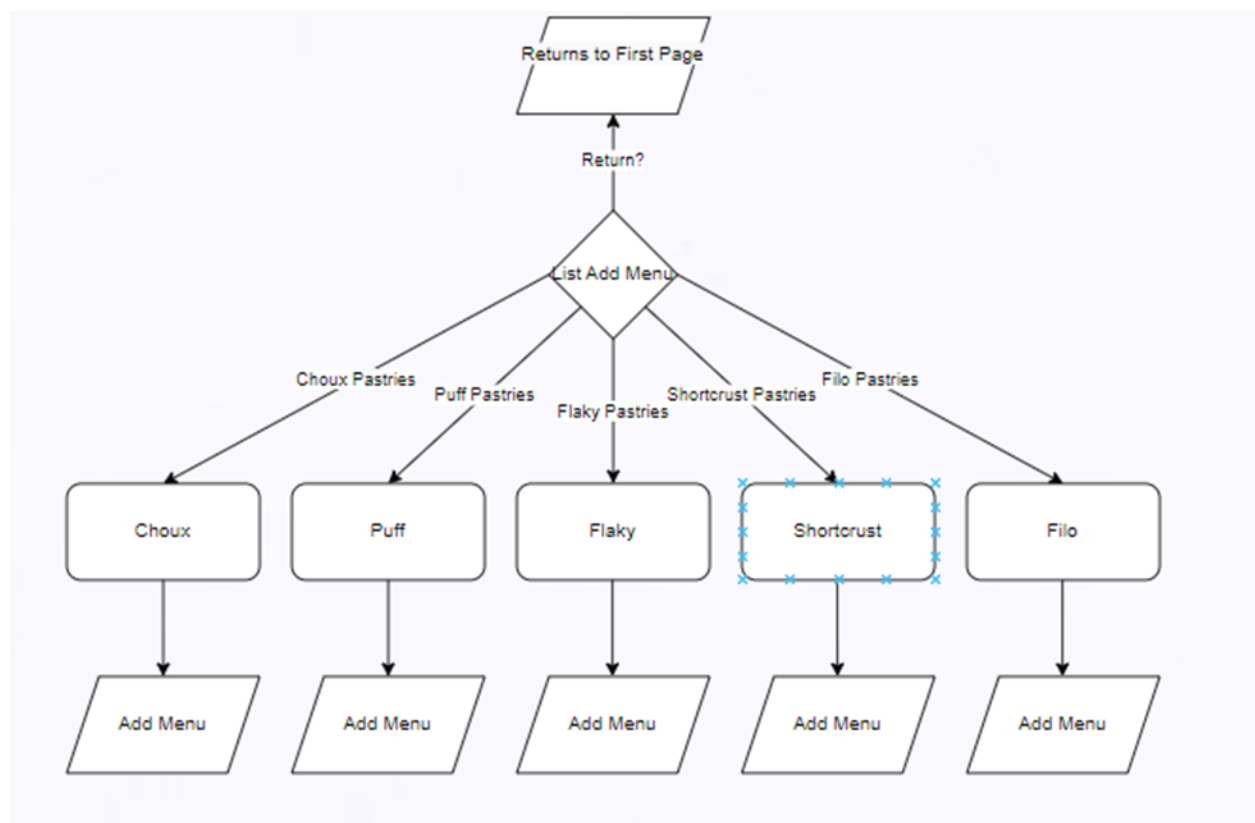


Pastry class calls more classes that read the contents of the pastry text file and returns it as a variable to display.

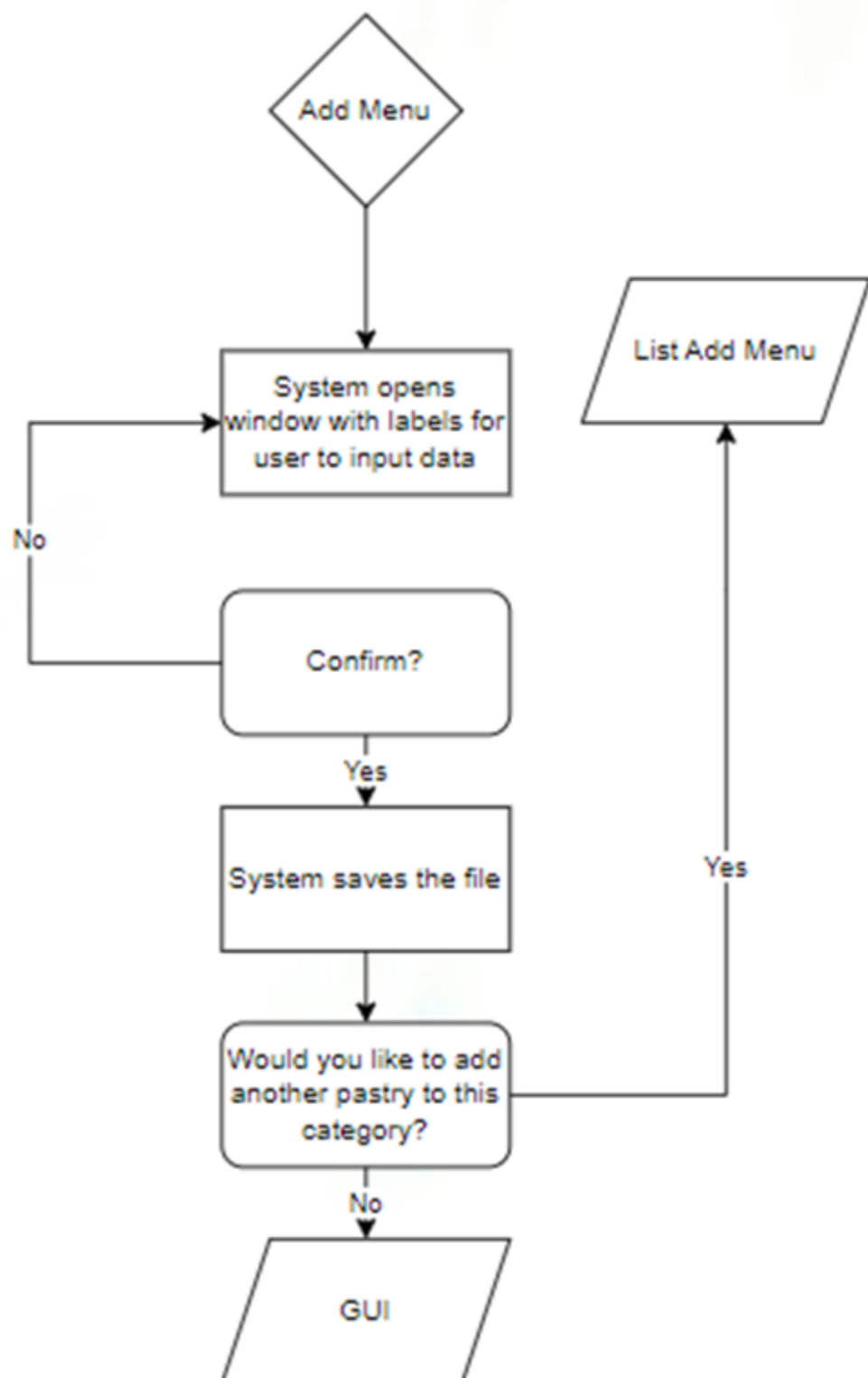


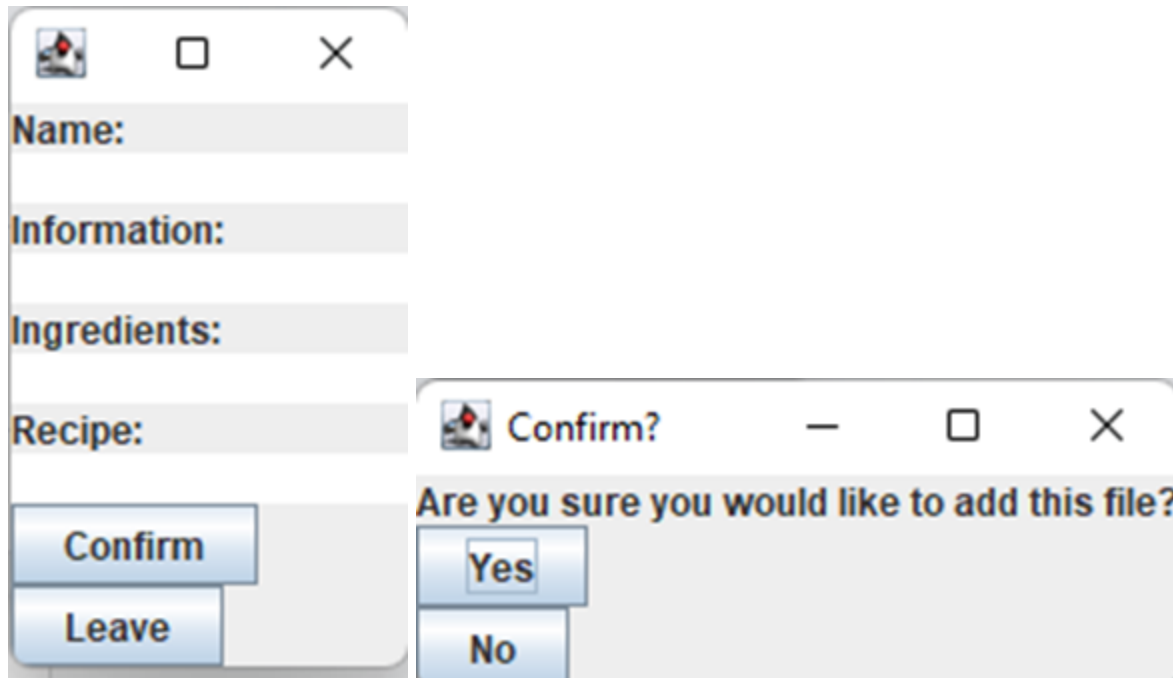


Flow Chart 3A:

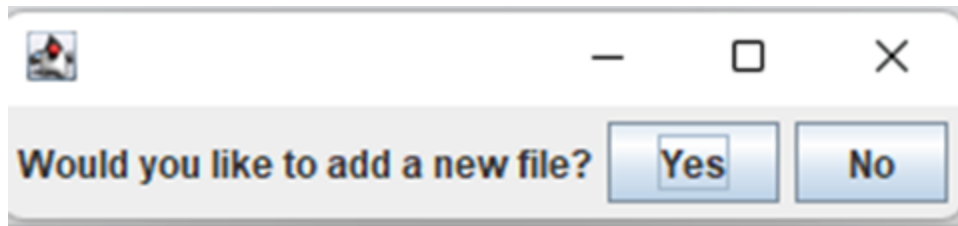


Flow Chart 3B:

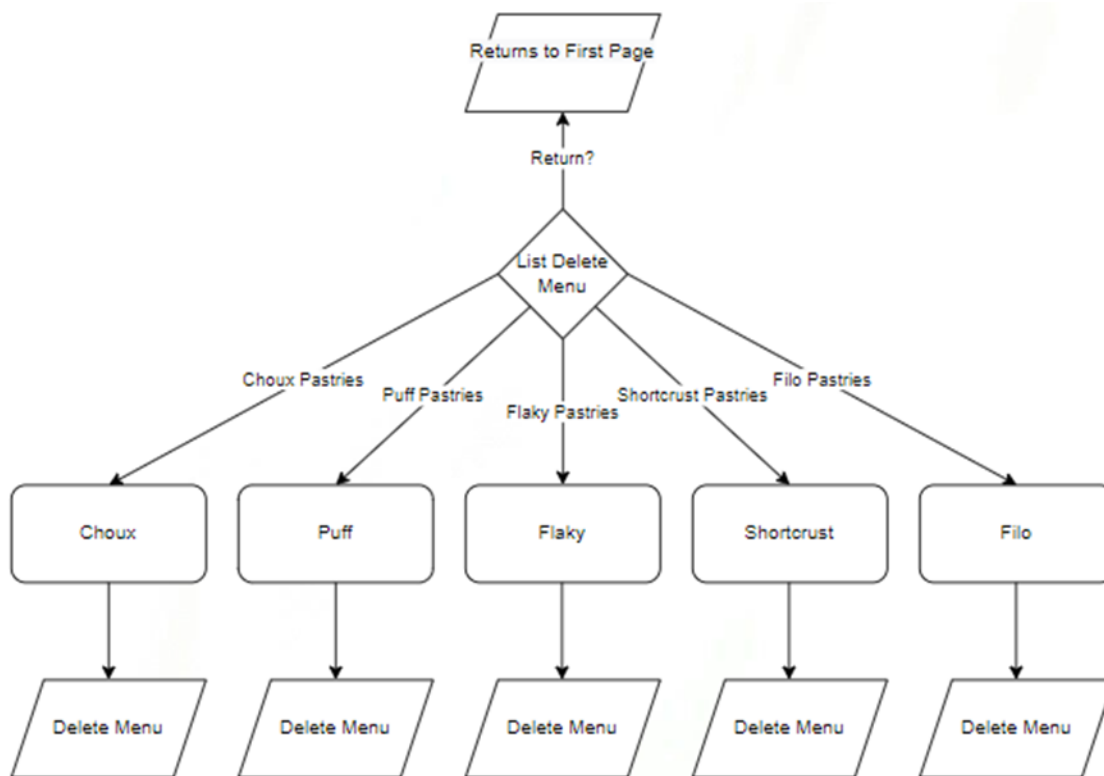




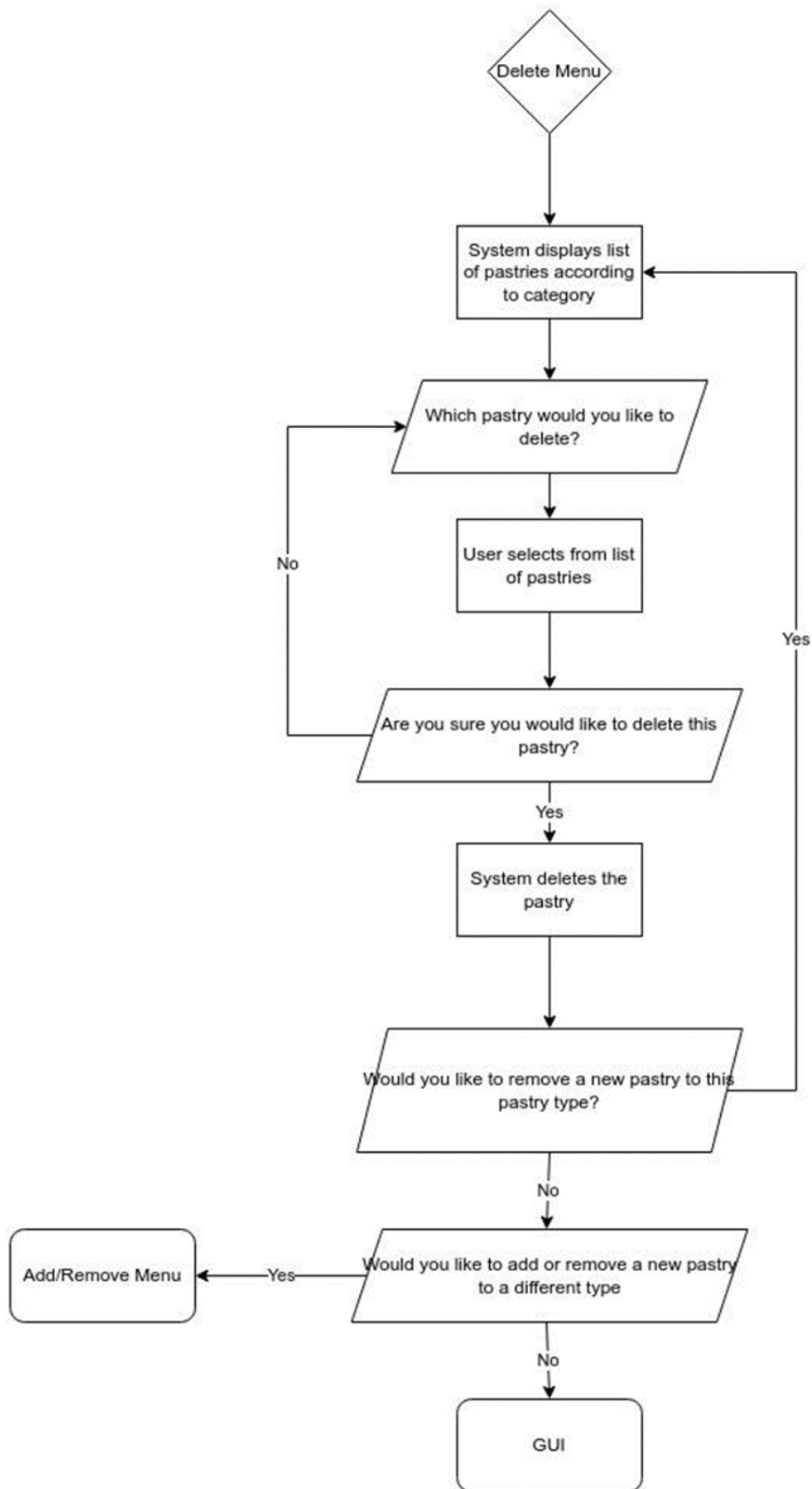
When the user clicks yes, the program checks if there is another file with the same name, if there is, it sends an error message and returns the user to the main menu. If there isn't a file with the same name, the program calls the filewriter module to write in what the user wrote and says it to the category that the user picks.



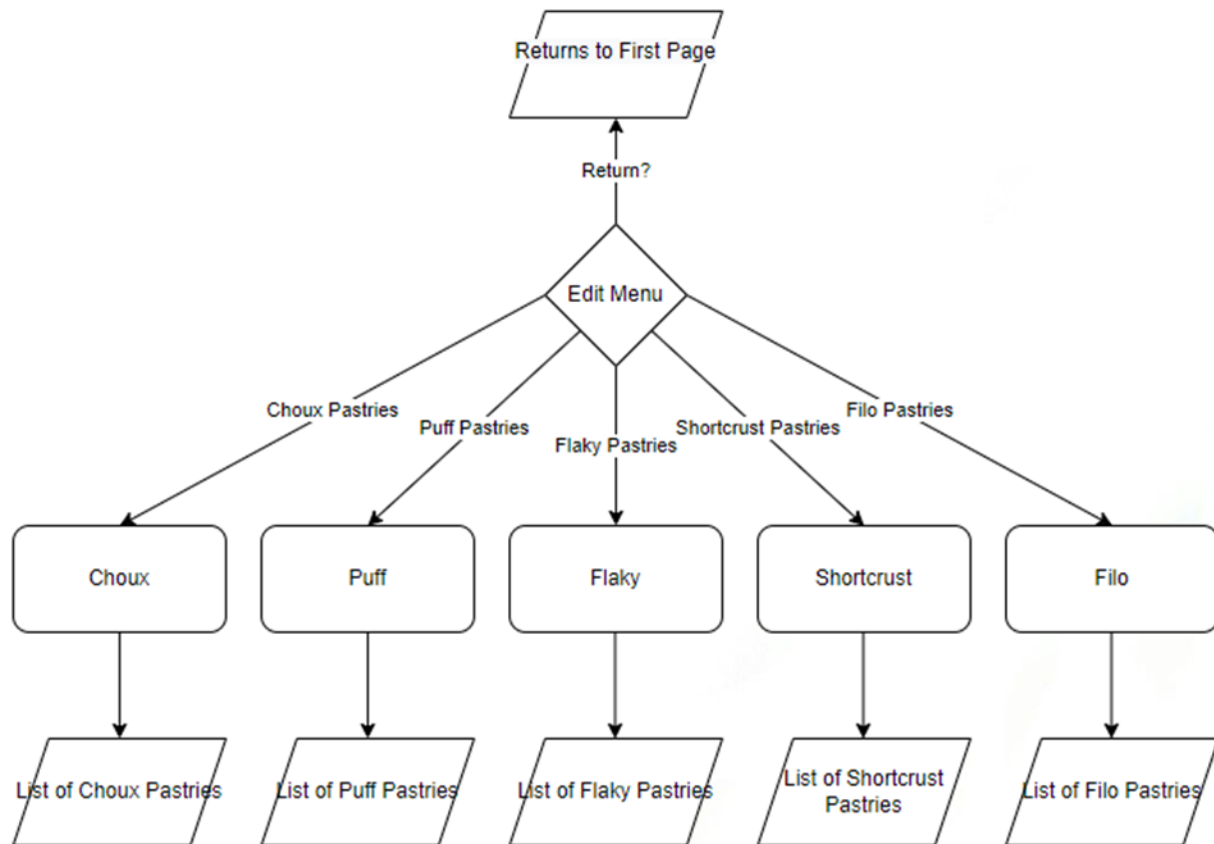
Flow Chart 4A:



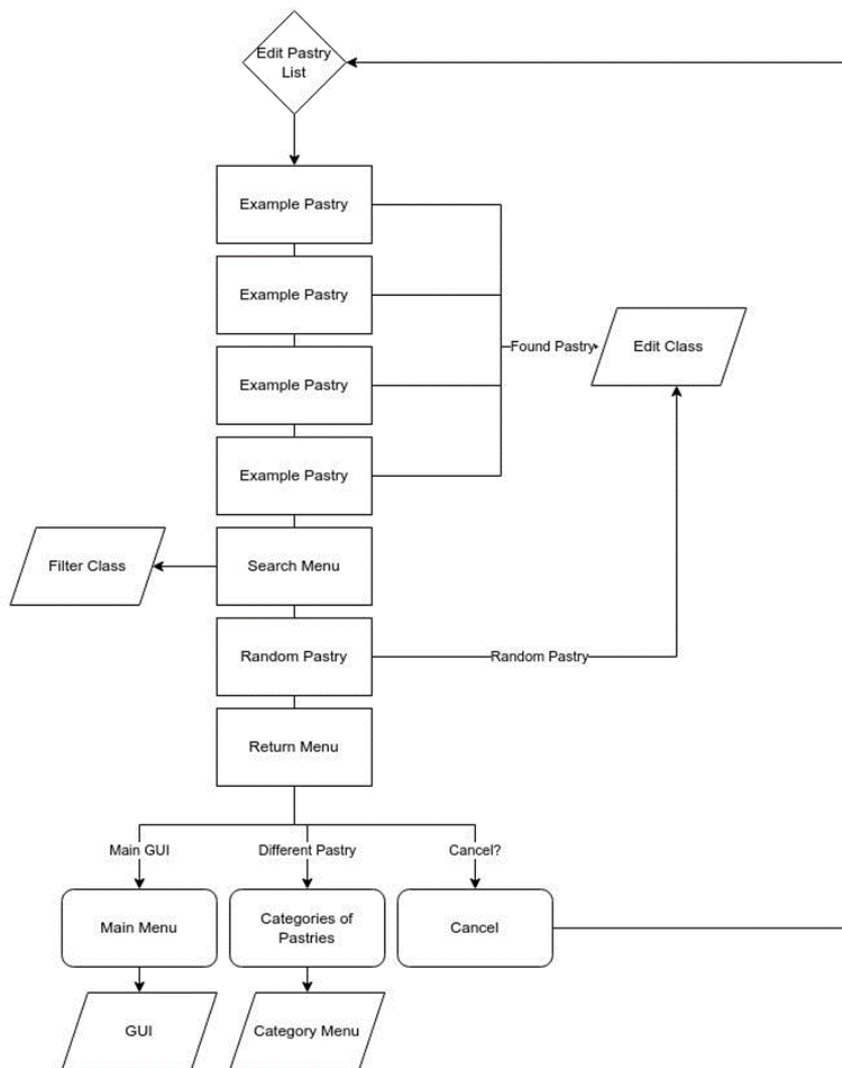
Flow Chart 4B:



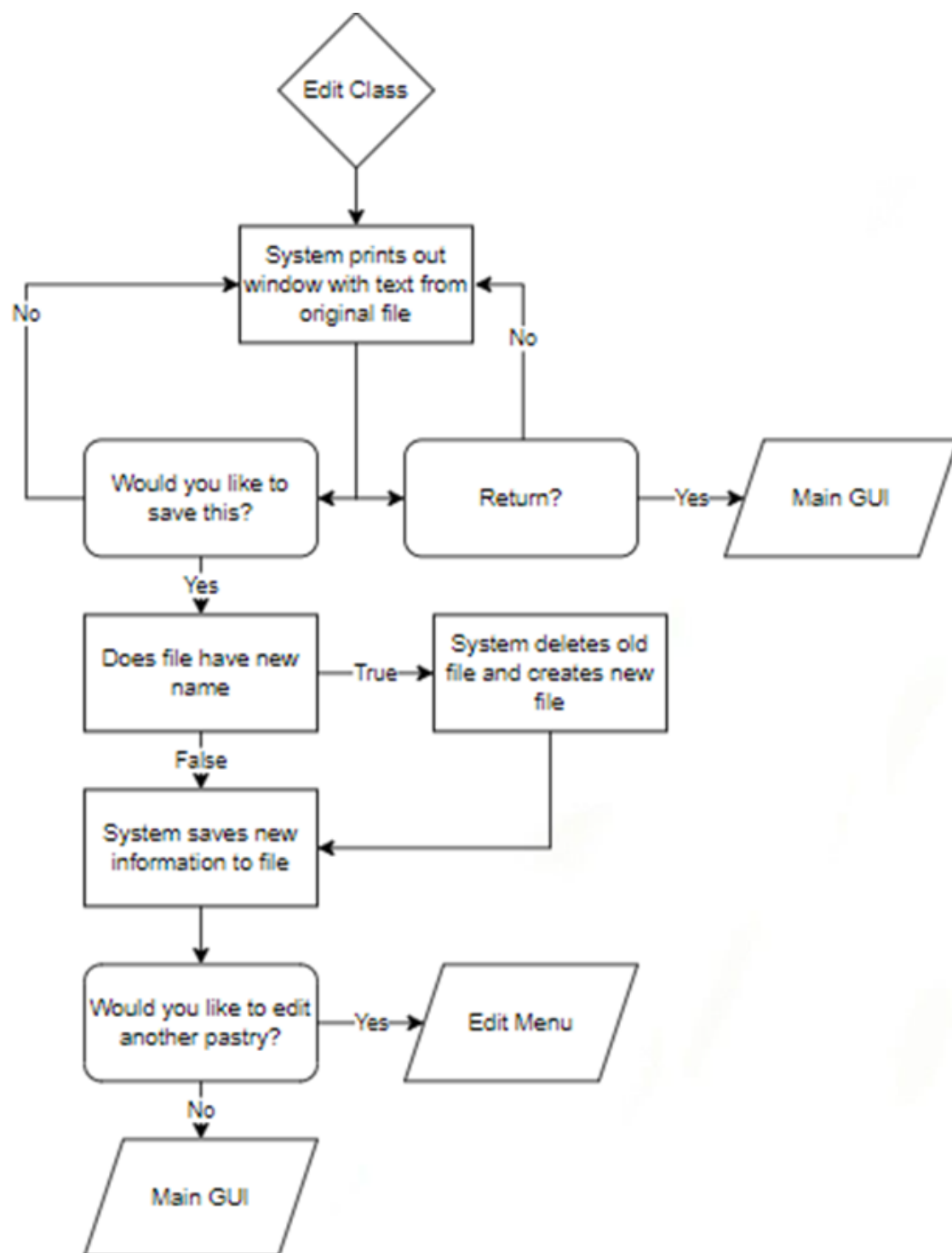
Flow Chart 5A:



Flow Chart 5B:



Flow Chart 5C:



Development Steps (Note the labels that reference the success criteria in criterion A)

1) Develop GUI with options - CS5

a) View different pastry - DB2

- i) Uses scanner class to read the file and puts text into an object of class

b) Add or remove - DB4

- i) Add: uses methods from File class to create a new file, uses FileWriter class to take inputted data and puts into the new file
- ii) Remove: Uses of File class's method: File.delete()

c) Edit - DB5

- i) Uses pastry's objects and inputs into JTextAreas, user can edit the text, after prompts to save, uses FileWriter class to modify the file

d) Quit

2) User input

a) Saving

- i) Create new file

b) Button creation - CS5

- i) Creates button variables presented by user

3) Add quit function

4) Create the categories

a) Main categories - CS1

- b) Display recipe - **DB3**
 - i) Takes objects from class and displays into JTextAreas
- 5) Develop method to add or remove - **DB4**
- 6) Develop search and sorting
 - a) Search for keywords
 - i) Recipe names - **CS3**
 - (1) Uses pattern class to compare input to pastry names
 - ii) Recipe ingredients - **CS2**
 - (1) Takes list of ingredients from object in pastry class, uses pattern class to compare ingredients to input
 - b) Sort alphabetically and reverse alphabetically
 - i) Uses methods from ArrayList to sort list of pastries
- 7) Develop editing system to edit - **DB5**
 - a) Edit recipe
- 8) Find a random recipe - **CS6**
 - a) Uses Random class to find a number of 1 to index of list

Testing Plan

Test Type	Nature of Test	Example
User input	Prompts and reacts to input correctly	Buttons on the GUI working properly
Adding to database	Create new pastries	Adding a recipe and saves to the database
Sorting and searching through database	Correctly sorts and finds the right elements of the user input	Sort by names/cost Search for names/ingredients
Editing pastries	Modifying the pastries	Altering parts of the recipes and the code removes original information to update with new information
Reading from the files	Check that the program can properly read specific elements of a pastry	Program displays all parts of file
Readability of program	GUI is easy to read for user	Labels for each object of pastry
Correct User Input	Determines if input is within the program	Displays error if user input is not in database

Return Menu	A GUI that navigates around the program based on prompt.	The menu takes the user to the right place
Random	Randomizes and picks a pastry	Running method multiple times to see if code randomizes